Software Engineering

CSC648/848

Henry
VILLAR

USABILITY TESTING

Focus Groups

Important

Usability Testing and Evaluation

- Usability testing answers the question: is the product usable – mostly YES/NO, *rarely provides design suggestions*.
- To develop usable product need to apply UCD (User Centered Design) from day one, usability testing only verifies the usability
- *QA is different from usability testing*: QA checks if the product performs according to specs (functionality, performance, security, reliability, free of bugs etc.). <u>Usually done before final usability testing</u>

- Note: product can perform per specs, pass the QA but still not be usable and vice versa!

Important

Usability testing vs. focus groups

- Classic *usability testing* (testers run test plans and answer questionnaires) usually does NOT offer guidance on how to improve usability, it just validates “usable/not usable” and points to problems
- *Focus groups* help integrate some usability testing with design and improvement guidance – highly recommended.

Repeat often – follow UCD

Important

Usability testing and focus groups

- *Usability testing*: typical users given product and test cases; users test the system and answer questionnaire
  - Usually done toward the end of the cycle
  - Good for verification but gives little deign guidance how to improve

- *Focus groups (internal or external)* – combine usability testing AND design improvements: preferred in early stages
  - 3-8 people in the room observe the product together and try use cases
  - Discuss problems, come up with better design
  - Scribe records the decisions /tasks (usually a lot of small details)
  - Changes made – keep the list of tasks and what was changed
  - Repeat and iterate (internally then externally)
- Apply User Centered Design – do focus groups often starting early on design and implementation

Focus groups - Essential tool in developing requirements in UCD

Important

- http://en.wikipedia.org/wiki/Focus_group
  - "In usability engineering, a focus group is a survey method to collect the views of users on software or a website. This marketing method can be applied to computer products to better understand the motivations of users and their perception of the product. Unlike other methods of ergonomics, focus group implies several participants: users or future users of the application. The focus group can only collect subjective data, not objective data on the use of the application as the usability test for example."
  - Can be done using mockups and prototypes

Usability Metrics

Important

- The International Organization for Standardization (ISO) recommends that usability metrics should include:
  - **Effectiveness:** The accuracy and completeness with which users achieve specified goals
  - **Efficiency:** The resources expended in relation to the accuracy and completeness with which users achieve goals.
  - **Satisfaction:** The comfort and acceptability of use.

Important

Usability Metrics: *Effectiveness*

- *First:* test that available functions allow completion of main tasks/use cases
- *Measure task completion*: % of people who completed the defined task in defined time
- *Count errors per task*

- *One suggested way to implement:*

Test/use case	% completed	errors	comments
search			
upload			
post			

Example of effectiveness issue

- Function: upload content and metadata
- Users: average computing skills
- GUI functionality – two options to evaluate:
  1. Upload text via WWW form

  OR

  2. Upload content via command line interface FTP

- Probably most will not be able to complete the test 2) although the function exists ➔ need to measure this

Important

Usability Metrics: *Efficiency*

- *Efficiency in time:*
  - average time to complete the task;
  - average time of those who completed the task/av. time of all users
- *Efficiency in effort*: e.g. number of clicks
- *Efficiency in content/design*:
  - number of screens;
  - Number of clicks
  - number of pages of instructions

(Note: many of these can be instrumented in the app and measured automatically)

Example of efficiency issues

- You need to annotate (view then assign tags) 500 images a day for experiment in Machine learning
- GUI solution A – total of 5 clicks
  - Open file
  - Chose file
  - Open and View image
  - Annotate
  - Save annotation
  - Go back to open file (but what was the last image you sued there?)
- GUI solution B – save 2 clicks, 10-15 sec. and less errors
  - Open and View image
  - Annotate
  - Save annotation and NEXT

  Savings: 500 images X 15 sec. = about 2 h/day!!!!!! Plus less fatigue and errors

Usability metrics: *Satisfaction (subjective)*

Important

- *Task satisfaction*:
  - Ask users to perform the task then run survey where user answers/rates how they "feel" about usability (e.g. Lickert scale questions)
  - Optional: allow users to enter comments

Satisfaction testing is the main subject of the subsequent slides

Example questionnaire – is it correct?

Question 15:

How easy was it to create VM?

Very difficult | Average | Very easy

Comments:

**** *This OK?* ****

<< Previous | Next >> | END

Example questionnaire CORRECTED

Important

Question 15:

It was easy to create VM

Statement, NOT a question

Strongly agree	Agree	Neutral	Disagree	Strongly disagree
●	○	○	○	○

Comments:

<< Previous

Next >>

END

User Centered Design and Usability testing

Important

- Recall UCD – involve users early and in all steps of design and development ➔ perform usability testing often and start early
  - Mockups and storyboard
  - “Horizontal prototypes” e.g. screens with no backend
  - Parts or whole product
  - APIs (even before implementation)
  - Final candidate products

Important

Usability Testing

- Process that employs representative target population participants to evaluate product usability using specific usability criteria
- Tests should cover key use cases

- Usability testing is not a guarantee for product success, but it should identify at least the key problems
- It answers the question "is the product usable" but usually does NOT offer guidance how to improve it (use focus groups for that)
- There are many cases where usability testing and focus groups were wrong!
- Good and practical book: J. Rubin: "Handbook of Usability testing", Wiley Technical Communication Library, 1994

Important

Typical usability test

- Chose *representative population*
- Get *legal agreements* in place
- Let testers run test as per *test plans* in chosen *test environment*
- Collect *questionnaires and measurements*
- *Analyze results* and present a *report* to developers, management etc.
- Developers; take action on the results – *revise, iterate and retest*

Types of usability tests

Exploratory:

- Early in the process
- Can be based on any form of the GUI (sketch, wire diagrams etc.)
- Evaluate preliminary, basic design concept
- Perform representative tasks in a "shallow" mode
- Informal test methodology, a lot of interaction
- Discuss high level concepts

Types of usability tests

Assessment

- Done After fundamental concepts are done
- Evaluates usability of lower level operations
- The users actually perform set of well defined tasks
- Less interaction with test monitor
- Quantitative measurements are collected

Types of usability tests

Validation

- Done late in development cycle, close to release
- Goal is to certify product usability, “disaster insurance” against launching a poor product
- Often the first time when the whole product si tested (including help and docs)
- Evaluate wrt. some predetermined usability standard or benchmark
- Standards come from previous testing, competitive information, marketing etc.
- Very specific quantitative tests
- Can establish standards for future products
- Can be also done by beta customers

Types of usability tests

Comparison

- Can be done at any point in the development cycle
- Compare alternatives using objective measures
- Can be informal of formal, depending when it is done
- Often, best of alternative designs is combined

Milestone 4: **Beta Launch, QA and Usability Testing and Final Commitment for Product Features (P1 list)**

Objectives:

- To make final commitment for functions to be delivered (e.g. list of Priority 1 functions) and which will be used in grading
- To check that all required non-functional specs are satisfied or on track
- To practice formal usability test plan development
- To practice formal QA
- To practice code review
- To ensure basic practices of secure SE are applied
- Ensure effective teamwork
- Ensure software development is effective
- Deliverable:
  - Document for review (around 8 pages)
    1. Product summary
    2. Usability test plan
    3. QA test plan
    4. Code Review
    5. Self-check on best practices for security
    6. Self-check: Adherence to original Non-functional specs

M4-5

- **Milestone 4.** Beta Launch, QA and Usability Testing and Final Commitment for Product Features (P1 list). November 12th 2024 (W12)

- **Milestone 5.** Final Demo and Documents for Grading. December 10th 2024 (W15)

Final Project Main Functions

- <u>Web App: Voting Guide with AI+</u>

1. **Search & Filter** by state, political party and issues/policies.
2. **Enter** official information (political parties, voter registration, polling location, voting process, etc.)
3. Site **Registration** (College Students, General public, Officials, Admin, etc.).
4. **Rate** candidates.
5. Search results may be displayed **using maps**
6. **AI capabilities** is a plus.

Two entrepreneurs, SFSU alumni, decided to create a start-up comprising of a team of about 6+ students to develop a web-based service to help USA college students and general public to identify *county, state, and federal*/national **candidates** running for public offices and their respective *policies*, and *background*. The web application will allow officials from the different **political parties** to enter the corresponding official information, and the **Elections office** to enter information about voter registration, polling locations, and voting process. College students and general public should be able to search by state, political party and issues/policies so results can list the corresponding data, and at the same time; user (student and general public) can enter their issues of interest and the AI chatbot can assist them with mapping what candidates are aligned with the user interest. This application is targeting three primary audiences: (1) College Students, (2) General Public, and (3) Officials from the political parties and election office. Students and general public users should be able to rate candidates. Officials from political parties and election office may need to be registered to use all the functionality of the application. General users may access to limited search capabilities and application functionality. Search results may display maps.

M3 Review (Week 10)

- **Milestone 3.** Review of functionality, UI, SW and planning for final product delivery
  - 07:30-07:50PM Team 6
  - 07:50-08:10PM Team 5
  - 08:10-08:30PM Team 4
  - 08:30-08:50PM Team 3
  - 08:50-09:10PM Team 2
  - 09:10-09:30PM Team 1

Part 1: Each team meets with Prof. Villar
Review of functionality, UI and general project status (Zoom)

1. *UI and functionality feedback (P1 functions only)*
2. *Brief review of code, github, database etc.*
3. *Project status – be ready to verbally explain status of the issues below*
4. *List of P1 features committed for delivery agreed at M3 review meeting*

Part 2: Review offline

- Send email with this data to ME and:
  - Durga Siva Lokesh Telaprolu (TA/Grader) at dtelaprolu@sfsu.edu

Appendix I – form to be filled out and included in e-mail body for M2 vertical prototype delivery

Milestone 2 Vertical Prototype Form
Team Number <N>

The below form is used for submission of needed information for the current milestone. The below table is used to help access certain parts of your web application. Please make sure the information submitted is accurate and up to date to ensure grading of the
milestone is completed in an efficient manner.

Item	Credentials
Website URL	
Website URL to search page	
SSH URL	
SSH Username	
SSH Password/Key	
Database URL	
Database Username	
Database Password	
Link to GitHub page that performs the Search	

Addition information, anything to help with executing or grading of the current milestone
Please explain what is working for your prototype and what can be tested. Also, what search terms to use. Please make sure to explain how to use certain pieces of the log in information. Especially if its more than a simple SSH command. If you server requires a Key please make sure to include it with the submission of the document, regardless if it was submitted with prior milestones. If you are worried about sending keys though emails , you can use links to google drive or drobox etc. Once key has been retrieved you can delete the link.

Study Materials

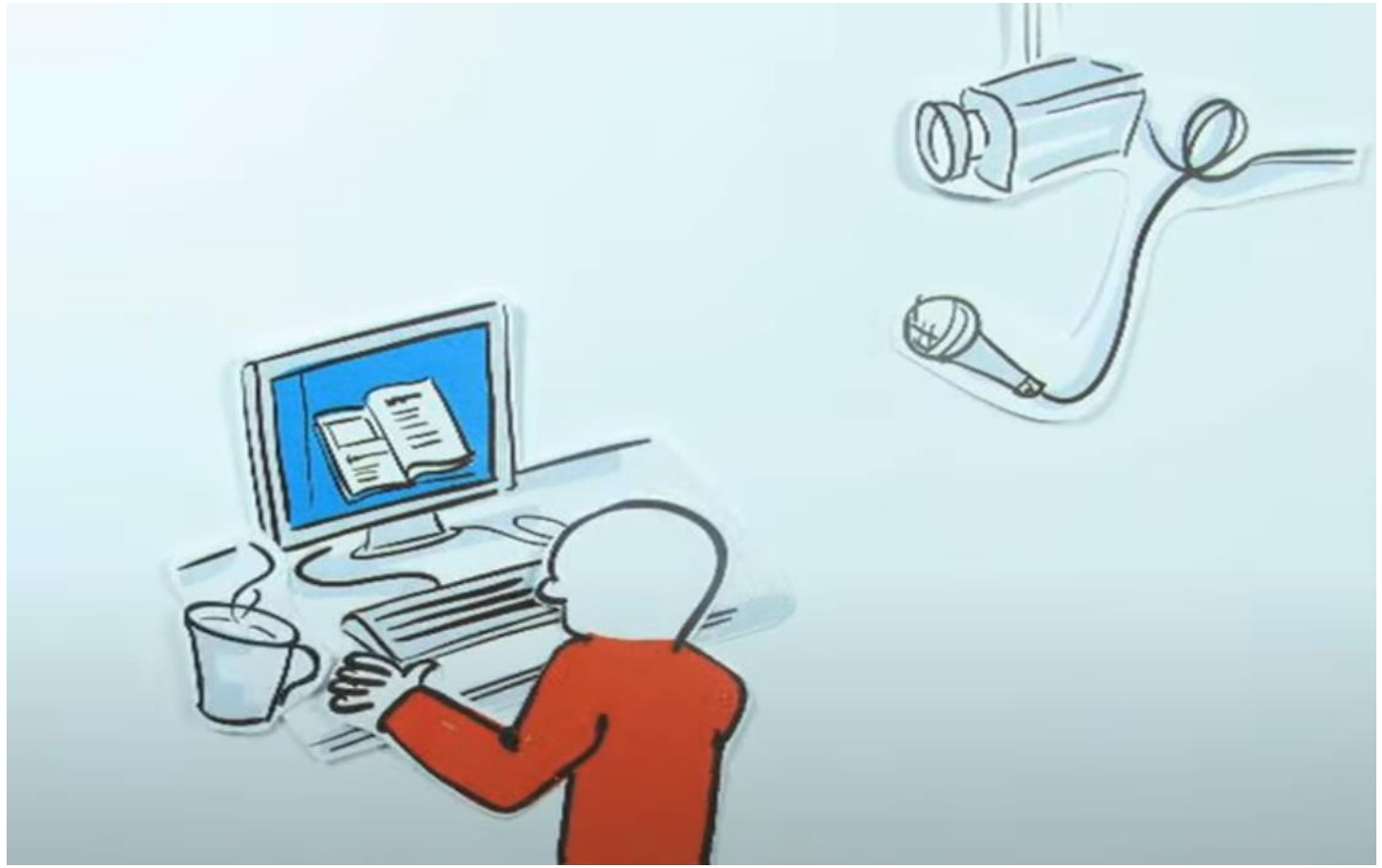
Can you do usability on low-budget? Yes, you can

- Make usability integral part of design and development.
- Start from the beginning! Use use cases….
- Involve graphics, pubs, sales, support, marketing, QA. They are your UCD Multidisciplinary team. They will not only help but feel they are part of the process, and also will be educated ahead of time about the new product.
- Use people in the company (but not from Engineering) and your friends
- Offer rewards (lunch etc.)
- Prioritize tests
- Prepare simple plans
- Get a quiet place and do it
- Be aware of the limitations on result quality
- Any approach is better than no usability testing

Usability metrics as per ISO – good overview

- http://ui-designer.net/usability/usersgoals.htm

Usability Testing - Video



https://www.youtube.com/watch?v=BrVnBdW6_rE

<u>Usability testing</u>: practical instructions

- We will present some practical and simple instructions for usability testing
- Note that for many culminating projects you will be required to do this
- Formal plan has to be submitted for review to SFSU Human Subject Protocol committee (Federal rules) IF results will be published

- Focus of subsequent slides is testing of *user satisfaction* (it is assumed that UI has been effective e.g. allows completion of key tasks)

Typical test plan format

Important

- Purpose: what is the main purpose of the test
- Problem statement: specific questions you want resolved
- Usability test/task plan :
  - describe actual tasks the user will perform – write it as instructions for the user
- User profile: who will be the users
- Method and test design: how will you observe it, how will you collect the data etc.
- Test environment and equipment
- Test monitor role
- Evaluation measures and data to be collected: how will you collect the feedback and how will you evaluate it
- Legal issues (voluntary, privacy)
- Report: what will final report contain

Task selection for evaluation

Important

- Tasks/tests to be evaluated are functions users want to do with the product - You then measure and evaluate user feedback
- Tasks/Tests: Cover only WHAT user has to do, NOT HOW to do it: Examples:
  - Create and file document
  - Find the right document
  - NOT: click on X, enter Y then click on Z…(this is QA not usability)
- Objective is to indirectly expose usability flaws by asking the user to perform typical tasks and NOT telling them exactly how to do it.
- Choose key and most frequently done tasks covering key use cases
- The task has to be specific and measurable (quantitatively or qualitatively)

Important

Task description

- Correct:
  - Do X (I.e. load paper)
- Not correct:
  - Do X by clicking here, then using this, then typing that…
- It is deliberately left to the user to figure out HOW to accomplish the task and to grade ease of use, intuitiveness etc.

Do NOT tell them how! Important

Task description

TASK	DESCRIPTION
Task	Load paper into copier
Machine state	Paper tray empty
Successful completion criteria	Paper properly loaded
Benchmark	Completed in 1 minute

Task components

TASK	DESCRIPTION
Task	Create VM using New VM Wizard
Machine state	VMware WS SW just loaded
Successful completion criteria	Working VM created
Benchmark	Completed in 30 sec.

Selection of evaluators and test groups

Important

- Evaluators should be representative of the targeted users – use your **personas** specification
- Independent groups or within-subject design (but be careful to avoid exposing users to same tests since this will bias the results)
- Adequate numbers of testers
- Offer motivation and rewards but NOT bribes
- Testers have to be informed of rules and sign the legal documents when required

Test environments

- Simple single room setups
  - Observer/monitor close to evaluator
  - Observer removed from evaluator
- Electronic observation room
- Classic elaborate usability lab
- Mobile lab

Also:

- Instrument the app as much as possible to measure:
  - Task completion efficiency (clicks, screens, time)
  - Errors

Important

Legal, privacy issues

- Serious legal and privacy issues related to managing the testers. Usually, tests have to be:
  - Voluntary
  - Anonymous
  - Can be abandoned at any time
- US Government regulates “human subject testing”
- Any grant funded work or work to be published at SFSU and other academic institutions has the have usability testing (e.g. “ human subjects”) protocol approved prior to commencement of the work

Important

Questionnaires (for user satisfaction testing)

Lickert scale

- I found GUI easy to use (check one)

  __ Strongly disagree __ Disagree
  __ Neither agree or disagree
  __ Agree __ Strongly agree
  (can also assign numbers from –2 to 2)

Semantic differentials

- I found File Open menu (circle one)

  Simple 3 2 1 0 1 2 3 Complex

Has to be a statement
NOT a question

Important

Questionnaires

Fill in questions

- I found the following aspects of GUI particularly easy to use (list 0-4 aspects)

  _____

  _____

  _____

  _____

Important

Questionnaires

Check-box

- Please check the statement that best describes your usage of spell check

  --- I always use spell check

  --- I use spell check only when I have to

  --- I never use spell check

Important

Questionnaires

Branching questions

- Would you rather use advanced search

  --- NO (skip to question 19)

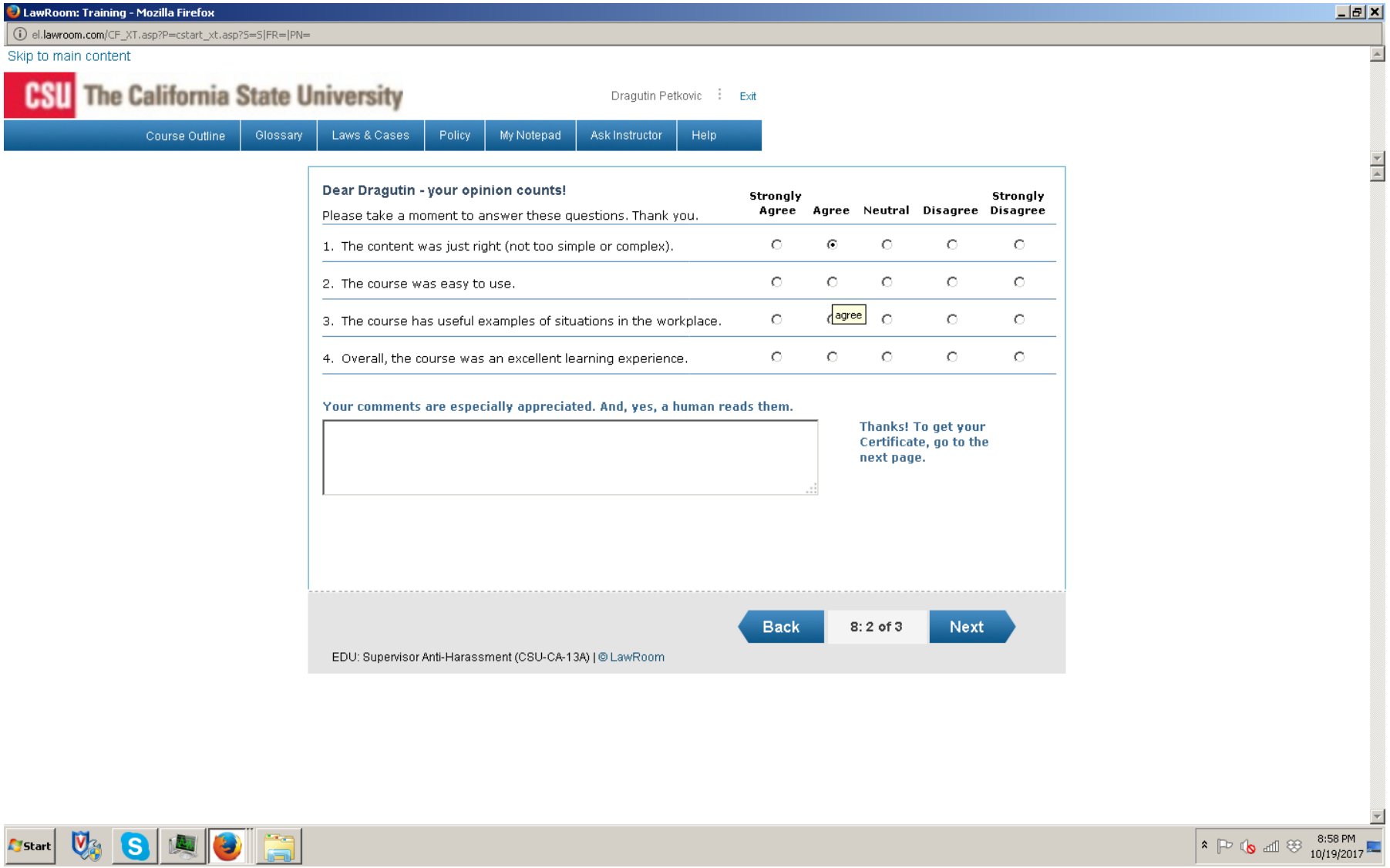
  --- YES (continue)

- What kind of advanced search would you like? (check one)

  --- Boolean

  --- Relevance

Example evaluation of on-line test



Important

Summarizing preference results

- Limited choice questions
  - Count how many participants selected each choice (number and %) – use bar charts
  - For Likert scale or semantic differentials provide
    - average and standard deviation of scores if there are enough evaluators
    - Histogram of user responses for each of 5 graded scale answers
    - Show % of testers who strongly agree/agree
- Free form questions
  - List questions and group answers into categories, also into positive and negative answers
- Free comments
  - List and group them at the end of the report. Options
    - By functions
    - Positive vs. negative

Hedgehog text annotations
- example of usability results (S. Raghavendra)

Statistical results:

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Process of adding annotations is easy and intuitive	0	0	1	4	1
The process of adding annotations is efficient	0	0	2	3	1
In real application, I would use annotation	0	0	2	3	1
Process of searching annotations is easy and intuitive	0	1	2	2	1
Searching annotations is efficient	0	1	1	3	1
In real application you would use the search function	0	1	1	3	1
Process of viewing annotations is easy and intuitive	0	0	0	3	3
Viewing annotations is efficient	0	0	1	2	3
	Too Clumsy	Difficult to work with	Very Intuitive		
Rate the look and feel of the user interface that you used to add, view and search annotations	1	1	4		

Important

Analyzing Data – high level summaries

- Identify and focus on tasks that did not pass the test or showed significant problems
- Identify user errors and difficulties
- Identify sources of errors
- Prioritize problems by criticality = severity AND probability of occurrence
- Analyze differences between groups (if applicable)

- Provide recommendations and prioritization at the end

Important Resources re: focus groups

Important

- About
  - https://en.wikipedia.org/wiki/Focus_group
  - https://www.nngroup.com/articles/focus-groups/

- How to run focus groups
  - https://www.interaction-design.org/literature/article/how-to-conduct-focus-groups
  - http://www.cse.lehigh.edu/~glennb/mm/FocusGroups.htm
  - https://uxdesign.cc/asking-the-right-questions-on-user-research-interviews-and-testing-427261742a67

Important

Future: UX testing in the era of on-line services and big data

- Products not released is well orchestrated downloads but continuously available and modified (e.g. on-line services, on-line search)
- UX testing is now augmented by “test-in-production” where UX testing is being continuously done from real customer usage
  - Data scientists and ML experts analyze and discover/predict user patterns and behaviors ➔ products are redesigned

Thank You